

■ Bems Farms — Driver Mobile App API Specification

Official Engineering Guide for Mobile App Developers (React Native, Flutter, iOS & Android)

Target Mobile Platforms: React Native / Flutter / iOS / Android	Authentication: JWT Bearer (`Authorization: Bearer <token>`)
Production API URL: https://api.bemsfarms.com/api/driver	Local Development URL: http://localhost:5000/api/driver
Payload Standard: application/json	Driver Wallet & DVA: Monnify Virtual NUBAN + Zone-Based Earnings

1. Driver API Directory & Endpoint Index

Category	Method & Endpoint	Auth	Description & Mobile Trigger
Onboarding	GET /onboarding/verify?token=...	Public	Verify invite token & pre-populate driver signup details
Onboarding	POST /onboarding/submit	Public	Submit driver KYC, NIN, vehicle details & settlement bank
Auth	POST /auth/login	Public	Driver email/phone + password authentication & JWT token
Auth	POST /auth/forgot-password	Public	Send password reset token via SMS / Email
Auth	POST /auth/reset-password	Public	Confirm reset token and configure new driver password
Profile	GET /auth/me	Bearer	Retrieve driver profile, rating, stats & Monnify Virtual Account
Profile	PATCH /auth/profile	Bearer	Update driver phone number, profile photo, or bank details
Availability	PATCH /availability	Bearer	Toggle duty state: Online (ready for dispatch) vs Offline
Deliveries	GET /deliveries	Bearer	Fetch assigned active deliveries with customer addresses & items
Deliveries	GET /deliveries/:orderId	Bearer	Fetch full itemized breakdown for a specific delivery drop
Milestones	PATCH /deliveries/:orderId/status	Bearer	Progress milestone (accepted, en_route, arrived, delivered)
History	GET /deliveries/history	Bearer	Fetch paginated past delivery drops and completed trips
Telemetry	POST /location	Bearer	Background GPS coordinate stream ping (every 10–15s)
Wallet	GET /earnings	Bearer	Live wallet balance, Monnify DVA NUBAN, zone rate card & history
Wallet	POST /withdraw	Bearer	Request payout of unwithdrawn earnings to personal bank account

2. Driver Authentication, Onboarding & Profile APIs

2.1 Driver Login (`POST /api/driver/auth/login`)

Authenticates driver by phone number or email and password. Returns JWT token and driver profile with Dedicated Virtual Account (DVA).

Request Payload	Success Response (200 OK)
<pre>{ "emailOrPhone": "08123456789", "password": "Password123!" }</pre>	<pre>{ "token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...", "message": "Login successful", "driver": { "id": 6, "name": "Victor Kalu", "phone": "08123456789", "email": "victor.kalu@bemsfarms.com", "wallet_account_number": "3397202109", "wallet_bank_name": "Monnify / Wema Bank", "wallet_account_name": "BEM - Victor Kalu", "bank_name": "GTBank", "account_number": "0123456789", "wallet_balance": 16500.00, "is_available": true } }</pre>

2.2 Toggle Availability State (`PATCH /api/driver/availability`)

Request Payload	Success Response (200 OK)
<pre>{ "is_available": true, "latitude": 9.0765, "longitude": 7.3986 }</pre>	<pre>{ "message": "Duty status updated to Online", "is_available": true, "status": "active" }</pre>

3. Dispatch Deliveries & Milestone Progression

Drivers receive live delivery drops and progress them through 5 status milestones.

Milestone	Action Description	Customer Tracking State	Financial Trigger
accepted	Driver accepts delivery assignment	driver_assigned	—
awaiting_pickup	Driver at hub/store loading goods	packed_ready	—
en_route	Driver departed with order package	out_for_delivery	—
arrived	Driver arrived at customer destination	driver_arrived	—
delivered	Handover complete with photo proof	delivered	Auto-Credit Zone Commission
failed	Delivery attempted but customer unavailable	delivery_attempted	—

3.1 Update Milestone (`PATCH /api/driver/deliveries/:orderId/status`)

Request Payload	Success Response (200 OK)
<pre>{ "status": "delivered", "proof_note": "Handed directly to customer Mrs. Ngozi.", "proof_photo": "https://storage.googleapis.com/bems-uploads/proof_del_998.jpg" }</pre>	<pre>{ "message": "Delivery completed successfully", "delivery_status": "delivered", "order_status": "delivered", "commission_earned": 1750.00, "new_wallet_balance": 18250.00 }</pre>

4. Driver Wallet, Dedicated Virtual Account (DVA) & Withdrawals

4.1 Get Driver Wallet & Earnings (`GET /api/driver/earnings`)

Returns live wallet balance, unwithdrawn earnings, pending payout amounts, Monnify Dedicated Virtual Account (DVA), zone rate card, and recent transaction history.

```
{
  "wallet": {
    "total_earned": 28500.00,
    "total_paid": 12000.00,
    "pending_payouts": 0.00,
    "available_balance": 16500.00,
    "commission_per_delivery": 1750.00,
    "dedicated_virtual_account": {
      "account_number": "8559127267",
      "bank_name": "Monnify / Wema Bank",
      "account_name": "BEMS - VICTOR KALU",
      "is_frozen": false
    },
    "settlement_bank": {
      "bank_name": "GTBank",
      "account_number": "0123456789",
      "account_name": "Victor Kalu"
    }
  },
  "zone_rates": [
    { "zone_id": "ZONE001", "zone_name": "Zone 1 - Maitama & Asokoro", "driver_earning_fee": 700.00, "commission_percent": 70, "estimated_eta": "20 - 35 mins" },
    { "zone_id": "ZONE002", "zone_name": "Zone 2 - Garki, Wuse, Central Area", "driver_earning_fee": 1750.00, "commission_percent": 70, "estimated_eta": "25 - 45 mins" },
    { "zone_id": "ZONE003", "zone_name": "Zone 3 - Utako, Jabi, Mabushi", "driver_earning_fee": 2450.00, "commission_percent": 70, "estimated_eta": "30 - 50 mins" },
    { "zone_id": "ZONE004", "zone_name": "Zone 4 - Gwarinpa, Kubwa", "driver_earning_fee": 3500.00, "commission_percent": 70, "estimated_eta": "40 - 65 mins" },
    { "zone_id": "ZONE005", "zone_name": "Zone 5 - Lugbe, Airport Road", "driver_earning_fee": 5600.00, "commission_percent": 70, "estimated_eta": "45 - 75 mins" },
  ]
}
```

```
{
  "zone_id": "ZONE006", "zone_name": "Zone 6 - Interstate Express", "driver_earning_fee": 24500.00, "commission_percent": 70, "estimated_eta": "Same Day"
},
{
  "recent_commissions": [
    { "id": 104, "total_earned": 1750.00, "deliveries": 1, "status": "pending", "created_at": "2026-09-21T09:10:00Z" }
  ],
  "recent_payouts": [
    { "payout_ref": "PAY-MNFY-8819", "amount": 12000.00, "status": "paid", "processed_at": "2026-09-20T14:30:00Z" }
  ]
}
```

4.2 Request Withdrawal (POST /api/driver/withdraw)

Request Payload	Success Response (200 OK)
<pre>{ "amount": 10000.00, "bank_name": "GTBank", "account_number": "0123456789", "account_name": "Victor Kalu", "notes": "Weekly earnings payout" }</pre>	<pre>{ "message": "Withdrawal request of ₦10,000.00 submitted successfully!", "payout_ref": "PAY-MNFY-99201", "available_balance": 6500.00, "status": "pending" }</pre>

5. Zone-Based Compensation Rate Schedule (Zones 1–6)

Zone ID	Zone Coverage Area	Customer Delivery Fee	Driver Drop Payout (70%)	Estimated ETA
ZONE001	Zone 1 - Maitama & Asokoro	₦1,000.00	₦700.00	20 - 35 mins
ZONE002	Zone 2 - Garki, Wuse, Central Area	₦2,500.00	₦1,750.00	25 - 45 mins
ZONE003	Zone 3 - Utako, Jabi, Mabushi	₦3,500.00	₦2,450.00	30 - 50 mins
ZONE004	Zone 4 - Gwarinpa, Kubwa Expressway	₦5,000.00	₦3,500.00	40 - 65 mins
ZONE005	Zone 5 - Lugbe, Airport Road	₦8,000.00	₦5,600.00	45 - 75 mins
ZONE006	Zone 6 - Interstate / Regional Express	₦35,000.00	₦24,500.00	Same Day / Next Day

6. Mobile Client Engineering Checklist

- **1. Secure Storage:** Store the JWT token securely using SecureStore (Expo) / EncryptedSharedPreferences (Android) / Keychain (iOS).
- **2. Background GPS Streaming:** When online (is_available: true), stream background coordinates to POST /api/driver/location every 10–15s.
- **3. Navigation Deep Linking:** Deep-link customer coordinates to Google Maps (geo:\${lat},\${lng}?q=\${lat},\${lng}) or Apple Maps (maps://?daddr=\${lat},\${lng}).
- **4. Direct Customer Contact:** Provide 1-tap dialer (tel:\${phone}) and WhatsApp quick chat (https://wa.me/234\${phone}).
- **5. Live Earnings View:** Show driver's real-time unwithdrawn earnings and Dedicated Virtual Account (DVA) directly on the app's wallet tab.
- **6. Offline Caching:** Cache active deliveries locally so drivers can view delivery addresses and items even during cellular network dips.